

LESSON 111 (0.5 GROUND and 1.3 DUAL)

Lesson Objective

During this lesson, the applicant will become familiar with the fundamentals of flight in the airplane. This will include demonstrating various turns, climbs, and descents with various power settings with minimal instructor guidance. This will solidify the applicant's understanding of how to enter and level off from climbs and descents as well as how to roll out from turns. The applicant will also review previously learned maneuvers for proficiency.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Ask the student if they have any questions about climbing, turning, and descending procedures.
- Student questions.

References

- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 03 - Basic Flight Maneuvers.
- Warrior Maneuvers Manual.

Instructor Guidance

- This is the first time the student applies climbs, descents, and turns in the actual aircraft after practicing them in the simulator. Expect a noticeable difference in performance — the real aircraft feels nothing like the AATD. Acknowledge this with the student upfront and frame it as a normal part of the learning process.
- During the ground portion, quiz the student on climbing, descending, and turning procedures from Lesson 110 before heading to the aircraft. If they cannot recall the entry, maintaining, and level-off sequence for each maneuver, address the gaps on the ground rather than in the air.
- The completion standard for this lesson requires minimal instructor guidance — resist the urge to over-coach. Give the student the opportunity to work through deviations on their own before intervening. Safety-critical errors should be corrected immediately, but minor altitude and heading deviations are learning opportunities.
- Trim remains the most common deficiency at this stage. If the student is muscling the aircraft rather than trimming, pause instruction on the current maneuver and address the habit directly. Do not allow the lesson to progress until trim usage is consistent.
- When leveling off from climbs and descents, watch for students who wait until they reach the target altitude before initiating the level-off. The 10% VSI anticipation habit must transfer from the simulator to the aircraft — reinforce it on every single level-off.

Lesson Content (Ground)

Review Fundamentals from Lesson 110

- Quiz student on climbing, descending, and turning procedures from previous AATD lesson.

Climbing

- Review how to enter a climb, maintain a climb, and level off from a climb.
- Climbs:
 - Entry:
 - Power.
 - Pitch.
 - Rudder.
 - Trim.
 - Maintaining:
 - Lookout.
 - Attitude indicator.
 - Airspeed indicator.
 - Altimeter.
 - HSI.
 - Maintain coordination as needed.
 - Leveling off:
 - Anticipate by 10% of VSI.
 - Pitch.
 - Power.
 - Trim.

Descending

- Descents:
 - Entry:
 - Power.
 - Pitch.
 - Rudder.
 - Trim.
 - Maintaining:
 - Lookout.
 - Attitude indicator.
 - Airspeed indicator.
 - Altimeter.
 - HSI.
 - Maintain coordination as needed.
 - Leveling off:
 - Anticipate by 10% of VSI.
 - Pitch.
 - Power.
 - Trim.

Turning

- Medium Level Turns:
 - Entry:

- Lookout.
- Bank - ailerons.
- Coordination - rudder.
- Back pressure - stabilator.
- Maintaining:
 - Lookout.
 - Attitude indicator.
 - Airspeed indicator.
 - Altimeter.
 - Coordination.
- Leveling off:
 - Anticipate by 1/2 bank angle.
 - Roll the wings level.

Lesson Content (Dual)

Checklist Procedures

- Re-emphasize the importance of using checklists.
- Have the student run through all the checklists.
- Assist if needed.
- Have the student go through the "passenger briefing" and the "engine fire during start brief".

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

Operation of the Communications and Navigation Units

- Ensure that the student understands how to properly use COMS and NAV units.
 - Points of Emphasis:
 - Equipment ON/OFF
 - Database Currency
 - Volume
 - Squelch
 - Standby/Active Frequencies
 - Switch between navigation sources
 - Other useful functions such as: NRST page and Direct-To

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Review Previous Exercises as Required

- Use the appropriate lesson plan depending on what needs to be reviewed.

Entering the Climb

- Full power.
- Right rudder.
- Set pitch attitude for the assigned airspeed (use visual references and attitude indicator).

Maintaining the Climb and Corrections

- Keep the desired pitch attitude using TRIM!
- Maintain rudder input as needed.
- Maintain heading using HSI and visual reference.
- Emphasize to level off every 500 feet to scan for traffic.

Climbing at Different Airspeeds

- $V_{X \text{ climb}}$: about 12.5 degrees
- $V_{Y \text{ climb}}$: about 10 degrees.
- Cruise climb: about 7.5 degrees.
 - Subject to change based on current conditions.

Leveling Off from a Climb

- Anticipate by using 10 percent of the vertical speed indicator.
- Lower the pitch to straight and level pitch attitude.
- Reduce the power to straight and level power setting (2300 RPM).
- TRIM!

Entering a Descent

- Set power to idle.
- Left rudder.
- Set pitch attitude for a glide descent.
 - Do not let the student chase the airspeed.

Maintaining a Descent and Corrections

- Keep the desired pitch attitude using TRIM!
- Maintain coordination via rudder input as needed.
- Maintain heading using visual reference and the HSI.

Leveling Off from a Descent

- Anticipate by using 10 percent of the vertical speed indicator.
- Increase the pitch to straight and level pitch attitude.
- Increase the power to straight and level power setting (2300 RPM).
- TRIM!

Entering Medium Level Turns

- Check for traffic!
- Bank using the ailerons and hold a 30-degree bank (show where that is on the attitude indicator).
- Maintain coordination using the rudder.
- Increase back pressure.
 - More bank → more horizontal component of lift → less vertical component of lift → more back pressure required.

Maintaining Medium Level Turns and Making Corrections

- Maintaining the turn:
 - Lookout.
 - Refer to attitude indicator, airspeed indicator and altimeter.
 - TRIM for the increased back pressure.
- Corrections:
 - Change bank angle as needed.
 - Hold the correction.

Rolling Out from Medium Level Turns

- Anticipate by ½ of the bank angle.
- Roll the wings level - smoothly.
- Decrease pitch attitude.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 112.
- Assign homework:
 - Warrior Maneuvers Manual
 - Read Chapter 8 of the Airplane Flying Handbook - Airport Traffic Patterns.
 - Read Chapter 7 of the Airplane Flying Handbook – Ground Reference Maneuvers.

Completion Standards

This lesson is complete when the applicant can:

1. Correctly perform previously learned maneuvers with minimal instructor guidance.
2. Demonstrate proper sequential checklist procedures for before and after start, before and after taxi, run-up, takeoff, climb and cruise without instructor guidance.
3. Demonstrate how to enter and level off from climbs and descents as well as how to enter and roll out from turns with minimal instructor guidance.
4. Control rate of descent when performing a glide descent.
5. Explain the effect of altitude and wind on glide distance.